

```

746
*****
*****
746! *****
747
748
*****
*****
748! *****
749
750
*****
*****
750! *****
751 20May09 When subdividing activity codes into pseudo codes to the
CARMM program, these are the
751! places where they also
752             will need to be added:
753             Below where %addActvCodeToMxMail is being
called in (appends the
753! psuedo code to the mix mail map)
754             Below where Proc format is formatting all
the activity codes
755             In the excel file (ActivityCodesMaster
03AUG09.xls)
756             In the sas code of "mapclassRecast.sas"
NOTE not needed for delivery
756! cost model, as not
757             using DMM field.
758             In the mixed mail/direct actv code match file
"MxMailCode.csv" - add new activity
758! codes for Std Mail
759             Be sure to use the correct field in ActivityCodeMaster -
NewCRAProd
760             select KL.*, ac.NewCRAProd as class2
761
*****
*****
761! *****
762 REMOVE COMMENT ON SELECT CASING DEFINITION SECTION TO USE FOR CASING
ONLY!!!
763
*****
*****
763! *****;
764
765 resetline;
1  options MPRINT SYMBOLGEN;
2
3  libname IOCS 'D:\ExternalSSD\Folder 19\FY25\SAS\SASLib';
NOTE: Libref IOCS was successfully assigned as follows:
      Engine:          V9
      Physical Name: D:\ExternalSSD\Folder 19\FY25\SAS\SASLib
4  %let pathProg = D:\ExternalSSD\Folder 19\FY25\SAS\SASInputFiles;

```

```

5      %let pathOut = D:\ExternalSSD\Folder 19\FY25\SAS\KLTables;
6      %let FY = 25;
7      %let FYData = 25;
8
9      %macro assignRouteGroup();
10         *** The following statement assigns route code groups
11         *** so that reports can be created separately for
12         *** these groups.;
13
14         *set rgroup for special runs, for delivery cost model (3
values);
15         *Put Letter Route types in RGroup 1, SPR in RGroup 2, 99 in
RGroup 3;
16
17         if '71' <= route <= '83' then RGroup=1;
18         else if '84' <= route <= '98' then RGroup=2;
19         else RGroup=3;
20
21     %mend;
22
23
24     data b;
25     set IOCS.PRCPubCL25;
26     title1 'FY 20&FY. Qtr A CARMM System';
27     /***** The following variables are updated using Cluster
dataset
27 ! *****/
28     /***** Updated by Dan Zhang on Nov 16, 2020
28 ! *****/
29     dollar = READCOST; /* Cost (in dollar) for each carrier
reading based on cluster
29 ! sampling method */
30     act1st = ' ';
31     act234 = ' ';
32     F9252 = CRAFTGRP; /* Craft Subscript: 5=Full-Time Carrier;
6=Part-Time Carrier */
33     f264 = WEIGHTCAG; /* CAG Group used in the Control Total
Dollar: A-H */
34
/*****
*****
34 ! *****/
35     *** Subdivide activity codes ***;
36
37     ***This section is used for FCSP by Indicia as requested
annually by PRC ***;
38
39     select (Q23E02);
40         when ('E')          indicia = "Stamped ";
41         when ('F')
42             select (q23E05);
43                 when ('A')          indicia = "PVI ";
44                 when ('B')          indicia = "Meter ";
45                 when ('C')          indicia = "IMI ";

```

```

46             otherwise          indicia = "Error    ";
47         end;
48         when ('G')              indicia = "Permit    ";
49         otherwise              indicia = "Other     ";
50     end;
51
52     if F262 = '1060' then do;          * Subdivide First-Class SP
Letters;
53         if Q16F05 NE 'Y'  and Q16B01C NE 'B' and Q16E01 NE 'C' then
F262 = '1060';
54         *non-collection costs go into 1060;
55         if Q16F05 = 'Y'  then F262 = '1061'; ** Handling Collection
Mail;
56         else
57             if Q16B01C = 'B' then F262 = '1061';  ** Carrier is on a
Collection Route;
58         else
59             if Q16E01 = 'C' then F262 = '1061';  ** AMS route number
indicates Collection Route;
60     end;
61     if f262 = '1061' then do;
62         if indicia = "Stamped  " then F262 = '1061';
63         else
64             if indicia = "Meter    " then F262 = '1062';
65             else
66                 if indicia = "PVI      " then F262 = '1063';
67                 else
68                     if indicia = "IMI      " then F262 = '1064';
69                     else
70                         if indicia = "Permit    " then F262 = '1066';
71                         else
72                             F262 = '1065';
73     end;
74
75         ** CARMM Edits;
76         if f264='K' then f9252='9';
77         if f261 ne '4';
78         if dollar=0 and q01=' ' then delete;
79         else do;
80             if f260 in ('84', '89') then f260='87';
81         end;
82         if f9252 in ('5','6'); **carrier craft;
83     run;

```

NOTE: Character values have been converted to numeric values at the places given by:

```

      (Line):(Column).
      76:32   82:12

```

NOTE: Variable q01 is uninitialized.

NOTE: There were 150303 observations read from the data set IOCS.PRCPUBCL25.

NOTE: The data set WORK.B has 102200 observations and 304 variables.

NOTE: DATA statement used (Total process time):

```

      real time          0.17 seconds

```

cpu time

0.09 seconds

SYMBOLGEN: Macro variable PATHPROG resolves to D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles

84

85 *Read in file that maps activity codes into mixed mail codes,
86 and add all new activity codes added above;

87 %include "&pathProg\macMxMail_AMS.sas" / source2; /*GET MORE DETAILS
IN THE LOG FOR THIS

87 ! INCLUDE STATEMENT*/

NOTE: %INCLUDE (level 1) file D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\macMxMail_AMS.sas

is file D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\macMxMail_AMS.sas.

88 +*Macros to read mixed mail activity code mapping, and insert new
activity codes;

89 +

90 +%macro readMxMail(CSVFile,MxMailMap);

91 +*Read file CSVFile (CSV format) with activity codes that belong to
mixed mail codes;

92 +* Prepare final mixed mail mapping dataset;

93 +Data mxmaill;

94 + infile &CSVfile DLM="," MISSOVER firstobs=2;

95 + format mixcode \$4.;

96 + format dircode \$4.;

97 + input mixcode \$ dircode \$;

98 +run;

99 +

100 +Data &MxMailMap; set MxMail1;

101 +run;

102 +%mend;

103 +

104 +%macro addActvCodeToMxMail(MxMailMap, actvMatch, actvNew);

105 +*insert new activity code actvNew with same mixed mail cosdes as
actvMatch;

106 +proc sql;

107 + insert into MxMailMap

108 + select mixcode, &actvNew

109 + from mxmaill

110 + where dircode = &actvMatch;

111 +quit;

112 +%mend;

113 +

114 +

115 +*Inserts activity code subsets that Nancy needs for processing,
these are not official

115!+activity codes;

116 +*AMS 17Nov17;

117 +%macro addActvCodeToMaster(ActivityCodesTable, ActivityCode,
NewCRAProd);

118 +*insert new activity code actvNew with same mixed mail cosdes as
actvMatch;

119 +proc sql;

```

120 + insert into ActivityCodesTable
121 + set ActivityCode = &ActivityCode.,
122 +     NewCRAProd = &NewCRAProd.;
123 +quit;
124 +%mend;
125 +
126 +
127 +/*
128 +%macro
addMixedMailShapeToMxMail(MxMailMailMap,mxMailNoShape,shapeCode);
129 +%Insert new mixed mail code with shape info, using first digit of
activity code
130 + as last digit of mixed mail code;
131 +proc sql;
132 +     insert into MxMailMap
133 +         select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
134 +     from mxMail1
135 +     where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
136 +quit;
137 +%mend;*/
138 +%macro addMixedMailShapeToMxMail(MxMailMailMap,mxMailNoShape);
139 +%Insert new mixed mail code with shape info, using first digit of
activity code
140 + as last digit of mixed mail code;
141 +%let shapeCode = 1;
142 +proc sql;
143 +     insert into MxMailMap
144 +         select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
145 +     from mxMail1
146 +     where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
147 +%let shapeCode = 2;
148 +proc sql;
149 +     insert into MxMailMap
150 +         select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
151 +     from mxMail1
152 +     where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
153 +%let shapeCode = 3;
154 +proc sql;
155 +     insert into MxMailMap
156 +         select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
157 +     from mxMail1
158 +     where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
159 +%let shapeCode = 4;
160 +proc sql;
161 +     insert into MxMailMap
162 +         select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
163 +     from mxMail1
164 +     where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) =
put(&shapeCode.,z1.);
165 +quit;

```

```

166 +%mend;
167 +
168 +
169 +/*
170 +*test macros;
171 +%let pathData = C:\AraSardar\IOCS\CARMM;
172 +%readMxMail("&pathData\MxMailCodeFY14.csv",MxMailMap)
173 +%addActvCodeToMxMail (MxMailMap,'1060','1061');
174 +%addActvCodeToMxMail (MxMailMap,'1060','1062');
175 +%addActvCodeToMxMail (MxMailMap,'1060','1063');
176 +%addActvCodeToMxMail (MxMailMap,'1310','1311');
177 +%addActvCodeToMxMail (MxMailMap,'1310','1312');
178 +%addActvCodeToMxMail (MxMailMap,'1310','1313');
179 +%addActvCodeToMxMail (MxMailMap,'1310','1314');
180 +%addActvCodeToMxMail (MxMailMap,'1310','1315');
181 +%addActvCodeToMxMail (MxMailMap,'1310','1316');
182 +%addActvCodeToMxMail (MxMailMap,'1310','1317');
183 +%addActvCodeToMxMail (MxMailMap,'1310','1318');
184 +%addActvCodeToMxMail (MxMailMap,'1310','1319');
185 +%addActvCodeToMxMail (MxMailMap,'2310','2311');
186 +%addActvCodeToMxMail (MxMailMap,'2310','2312');
187 +%addActvCodeToMxMail (MxMailMap,'2310','2313');
188 +%addActvCodeToMxMail (MxMailMap,'2310','2314');
189 +%addActvCodeToMxMail (MxMailMap,'2310','2315');
190 +%addActvCodeToMxMail (MxMailMap,'2310','2316');
191 +%addActvCodeToMxMail (MxMailMap,'2310','2317');
192 +%addActvCodeToMxMail (MxMailMap,'2310','2318');
193 +%addActvCodeToMxMail (MxMailMap,'2310','2319');
194 +%addActvCodeToMxMail (MxMailMap,'3310','3311');
195 +%addActvCodeToMxMail (MxMailMap,'3310','3312');
196 +%addActvCodeToMxMail (MxMailMap,'3310','3313');
197 +%addActvCodeToMxMail (MxMailMap,'3310','3314');
198 +%addActvCodeToMxMail (MxMailMap,'3310','3315');
199 +%addActvCodeToMxMail (MxMailMap,'3310','3316');
200 +%addActvCodeToMxMail (MxMailMap,'3310','3317');
201 +%addActvCodeToMxMail (MxMailMap,'3310','3318');
202 +%addActvCodeToMxMail (MxMailMap,'3310','3319');
203 +%addActvCodeToMxMail (MxMailMap,'4310','4311');
204 +%addActvCodeToMxMail (MxMailMap,'4310','4312');
205 +%addActvCodeToMxMail (MxMailMap,'4310','4313');
206 +%addActvCodeToMxMail (MxMailMap,'4310','4314');
207 +%addActvCodeToMxMail (MxMailMap,'4310','4315');
208 +%addActvCodeToMxMail (MxMailMap,'4310','4316');
209 +%addActvCodeToMxMail (MxMailMap,'4310','4317');
210 +%addActvCodeToMxMail (MxMailMap,'4310','4318');
211 +%addActvCodeToMxMail (MxMailMap,'4310','4319');
212 +proc sort data=MxMailMap nodup;
213 +by mixcode dircode;
214 +run;
215 +*/
216 +
217 +***** End macMxMail;
NOTE: %INCLUDE (level 1) ending.

```

```

SYMBOLGEN: Macro variable PATHPROG resolves to D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles
MPRINT(READMXMAIL): *Read file CSVFile (CSV format) with activity codes
that belong to mixed
mail codes;
MPRINT(READMXMAIL): * Prepare final mixed mail mapping dataset;
218 %readMxMail("&pathProg\MxMailCodeFY25.csv",MxMailMap); *for
domestic CRA;
MPRINT(READMXMAIL): Data mxmail1;
SYMBOLGEN: Macro variable CSVFILE resolves to "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv"
MPRINT(READMXMAIL): infile "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv"
DLM="," MISOVER firstobs=2;
MPRINT(READMXMAIL): format mixcode $4.;
MPRINT(READMXMAIL): format dircode $4.;
MPRINT(READMXMAIL): input mixcode $ dircode $;
MPRINT(READMXMAIL): run;

```

```

NOTE: The infile "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv" is:
      Filename=D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv,
      RECFM=V,LRECL=32767,File Size (bytes)=2624,
      Last Modified=13Nov2025:12:44:54,
      Create Time=17Nov2025:10:11:34

```

```

NOTE: 237 records were read from the infile "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv".
      The minimum record length was 9.
      The maximum record length was 9.
NOTE: The data set WORK.MXMAIL1 has 237 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time          0.02 seconds
      cpu time           0.00 seconds

```

```

SYMBOLGEN: Macro variable MXMAILMAP resolves to MxMailMap
MPRINT(READMXMAIL): Data MxMailMap;
MPRINT(READMXMAIL): set MxMail1;
MPRINT(READMXMAIL): run;

```

```

NOTE: There were 237 observations read from the data set WORK.MXMAIL1.
NOTE: The data set WORK.MXMAILMAP has 237 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds

```

```

219
MPRINT(ADDACTVCODETOMXMAIL): *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
220 *add subdivided activity codes to mixed mail mappings );

```

```

221
222  %addActvCodeToMxMail(MxMailMap,'1060','1061');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1061'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1061' from mxmaill where
dircode = '1060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds

MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
223  %addActvCodeToMxMail(MxMailMap,'1060','1062');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1062'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1062' from mxmaill where
dircode = '1060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds

MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
224  %addActvCodeToMxMail(MxMailMap,'1060','1063');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1063'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1063' from mxmaill where
dircode = '1060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds

MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as

```

```
actvMatch;
225 %addActvCodeToMxMail(MxMailMap,'1060','1064');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '1064'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '1060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1064' from mxmaill where
dircode = '1060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
226 %addActvCodeToMxMail(MxMailMap,'1060','1065');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '1065'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '1060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1065' from mxmaill where
dircode = '1060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
227 %addActvCodeToMxMail(MxMailMap,'1060','1066');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '1066'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '1060'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1066' from mxmaill where
dircode = '1060';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
```

```
actvMatch;
228 %addActvCodeToMxMail(MxMailMap,'1317','1315');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1315'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1315' from mxmaill where
dircode = '1317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
229 %addActvCodeToMxMail(MxMailMap,'1317','1316');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1316'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1316' from mxmaill where
dircode = '1317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
230 %addActvCodeToMxMail(MxMailMap,'1317','1318');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1318'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1318' from mxmaill where
dircode = '1317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
```

```
actvMatch;
231  %addActvCodeToMxMail(MxMailMap,'1317','1319');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1319'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'1319' from mxmaill where
dircode = '1317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
232  %addActvCodeToMxMail(MxMailMap,'2317','2315');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2315'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2315' from mxmaill where
dircode = '2317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
233  %addActvCodeToMxMail(MxMailMap,'2317','2316');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2316'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2316' from mxmaill where
dircode = '2317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
```

```
actvMatch;
234 %addActvCodeToMxMail(MxMailMap,'2317','2318');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '2318'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '2317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2318' from mxmaill where
dircode = '2317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
235 %addActvCodeToMxMail(MxMailMap,'2317','2319');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '2319'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '2317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'2319' from mxmaill where
dircode = '2317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
236 %addActvCodeToMxMail(MxMailMap,'3317','3315');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '3315'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '3317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'3315' from mxmaill where
dircode = '3317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
```

```
actvMatch;
237 %addActvCodeToMxMail(MxMailMap,'3317','3316');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '3316'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '3317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'3316' from mxmaill where
dircode = '3317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
238 %addActvCodeToMxMail(MxMailMap,'3317','3318');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '3318'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '3317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'3318' from mxmaill where
dircode = '3317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
239 %addActvCodeToMxMail(MxMailMap,'3317','3319');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '3319'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '3317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'3319' from mxmaill where
dircode = '3317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
```

```
actvMatch;
240 %addActvCodeToMxMail(MxMailMap,'4317','4315');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '4315'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '4317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'4315' from mxmaill where
dircode = '4317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
241 %addActvCodeToMxMail(MxMailMap,'4317','4316');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '4316'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '4317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'4316' from mxmaill where
dircode = '4317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
242 %addActvCodeToMxMail(MxMailMap,'4317','4318');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN: Macro variable ACTVNEW resolves to '4318'
SYMBOLGEN: Macro variable ACTVMATCH resolves to '4317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'4318' from mxmaill where
dircode = '4317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.
```

```
MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      cpu time           0.00 seconds
```

```
MPRINT(ADDACTVCODETOMXMAIL):  *insert new activity code actvNew with
same mixed mail cosdes as
```

```

actvMatch;
243 %addActvCodeToMxMail (MxMailMap, '4317', '4319');
MPRINT(ADDACTVCODETOMXMAIL):  proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '4319'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '4317'
MPRINT(ADDACTVCODETOMXMAIL):  insert into MxMailMap select mixcode,
'4319' from mxmaill where
dircode = '4317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

```

```

MPRINT(ADDACTVCODETOMXMAIL):  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds

```

```

244
245
246 proc sort data=MxMailMap nodup;
247         by dircode mixcode;
248 run;

```

```

NOTE: There were 311 observations read from the data set WORK.MXMAILMAP.
NOTE: 0 duplicate observations were deleted.
NOTE: The data set WORK.MXMAILMAP has 311 observations and 2 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time          0.01 seconds
      cpu time           0.00 seconds

```

```

249
250 data carr; set b;
251     Craft=F9252;
252     Actv=F262;
253     *Actv=F9806; *mcz 10Aug08 use F9806 to distribute Int'l mixed
mail;
254     Route=F260;
255     BF=F261;
256     Fin=F263;
257     Cag=F264;
258     Cagfin=cag !! fin;
259
260     if actv in ('5740', '5745', '5741') then actv = '5750'; /*as per
Audrey Hu, 17Nov23*/
261     if actv <= '0900' and actv not in ('0350', '0560') then actgrp=1;
262     else if '1020' <= actv <= '4910' or actv in ('0350', '0560')
then actgrp=2;
263     else if '5300' <= actv <= '5750' then actgrp=3;
264     else if '5040' <= actv <= '6720' then actgrp=4;
SYMBOLGEN:  Macro variable FY resolves to 25
265     TITLE1 "FY 20&fy. Qtr A CARMM SYSTEM";
266 run;

```

```

NOTE: There were 102200 observations read from the data set WORK.B.

```

NOTE: The data set WORK.CARR has 102200 observations and 312 variables.

NOTE: DATA statement used (Total process time):

real time	0.06 seconds
cpu time	0.06 seconds

```
267
268 proc sql;
269     create table smy as
270     select cag, fin, actv, bf, sum(dollar) as dollar
271     from carr
272     group by cag, fin, actv, bf;
NOTE: Table WORK.SMY created, with 394 rows and 5 columns.
```

```
273 quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.03 seconds
      cpu time           0.00 seconds
```

```
274
275
276 data smyx;
277     array bfx{4} bf1 - bf4;
278     do i=1 to 4;
279         set smy;
280         by cag fin actv bf;
281         if i=1 and bf='2' then i=i+1;
282         if i=1 and bf='3' then i=i+2;
283         if i=1 and bf='5' then i=i+3;
284         if i=2 and bf='3' then i=i+1;
285         if i=2 and bf='5' then i=i+2;
286         if i=3 and bf='5' then i=i+1;
287         bfx{i}=dollar;
288         if last.actv then i=4;
289     end;
290     if bf1=. then bf1=0;
291     if bf2=. then bf2=0;
292     if bf3=. then bf3=0;
293     if bf4=. then bf4=0;
294     total=sum(of bf1-bf4);
295     cagfin=cag !! fin;
296     if actv <= '0900' and actv not in ('0350','0560') then
actgrp=1;
297         else if '1020' <= actv <= '4910' or actv in ('0350','0560')
then actgrp=2;
298         else if '5300' <= actv <= '5750' then actgrp=3;
299         else if '5040' <= actv <= '6720' then actgrp=4;
300         output;
301 run;
```

NOTE: There were 394 observations read from the data set WORK.SMY.

NOTE: The data set WORK.SMYX has 384 observations and 13 variables.

NOTE: DATA statement used (Total process time):

```
real time          0.03 seconds
cpu time           0.01 seconds
```

```
302
303
304 proc sort data=smyx;
305     by cagfin actgrp actv;
306 run;
```

NOTE: There were 384 observations read from the data set WORK.SMYX.

NOTE: The data set WORK.SMYX has 384 observations and 13 variables.

NOTE: PROCEDURE SORT used (Total process time):

```
real time          0.01 seconds
cpu time           0.00 seconds
```

```
307 *-----;
308 *                                           ;
309 * Dollar tally summary by cag fin route actv bf. This      ;
310 * summary file is used in next step for distributing      ;
311 * mixed-mail.                                           ;
312 *                                           ;
313 *-----;
```

```
314
315 proc sql;
316     create table smy2 as
317     select cag, fin, route, actv, bf, sum(dollar) as dollar
318     from carr
319     group by cag, fin, route, actv, bf;
```

NOTE: Table WORK.SMY2 created, with 1526 rows and 6 columns.

```
320 quit;
```

NOTE: PROCEDURE SQL used (Total process time):

```
real time          0.03 seconds
cpu time           0.01 seconds
```

```
321
322
323 data smy2;
324     set smy2;
325
326 *Use macro to assign route group;
327     %assignRouteGroup;
MPRINT(ASSIGNROUTEGROUP):  *** The following statement assigns route
code groups *** so that
reports can be created separately for *** these groups.;
MPRINT(ASSIGNROUTEGROUP):  *set rgroup for special runs, for delivery
cost model (3 values);
MPRINT(ASSIGNROUTEGROUP):  *Put Letter Route types in RGroup 1, SPR in
RGroup 2, 99 in RGroup 3;
MPRINT(ASSIGNROUTEGROUP):  if '71' <= route <= '83' then RGroup=1;
MPRINT(ASSIGNROUTEGROUP):  else if '84' <= route <= '98' then RGroup=2;
```

```

MPRINT(ASSIGNROUTEGROUP):   else RGroup=3;
328       cagfin=cag !! fin;
329   run;

```

NOTE: There were 1526 observations read from the data set WORK.SMY2.

NOTE: The data set WORK.SMY2 has 1526 observations and 8 variables.

NOTE: DATA statement used (Total process time):

```

      real time           0.01 seconds
      cpu time            0.00 seconds

```

```

330
331   data out;
332       array bff{4} dollar1-dollar4;
333       do i=1 to 4;
334           set smy2;
335           by cag fin route actv bf;
336           if i=1 and bf='2' then i=i+1;
337           if i=1 and bf='3' then i=i+2;
338           if i=1 and bf='5' then i=i+3;
339           if i=2 and bf='3' then i=i+1;
340           if i=2 and bf='5' then i=i+2;
341           if i=3 and bf='5' then i=i+1;
342           bff{i}=dollar;
343           if last.actv then i=4;
344       end;
345       if dollar1=. then dollar1=0;
346       if dollar2=. then dollar2=0;
347       if dollar3=. then dollar3=0;
348       if dollar4=. then dollar4=0;
349       drop i cag fin _type_ _freq_ dollar bf;
350   output;
351   run;

```

WARNING: The variable _type_ in the DROP, KEEP, or RENAME list has never been referenced.

WARNING: The variable _freq_ in the DROP, KEEP, or RENAME list has never been referenced.

NOTE: There were 1526 observations read from the data set WORK.SMY2.

NOTE: The data set WORK.OUT has 1524 observations and 8 variables.

NOTE: DATA statement used (Total process time):

```

      real time           0.02 seconds
      cpu time            0.00 seconds

```

```

352
353
354   ****Note: this 2nd part of the program is analogous to the **;
355   ****      old program ALA860C0, which performed distribution **;
356   ****      and produced various summary reports.          **;
357   ****
358   ****
359
360   data out;

```

```

361      set out;
362      if '5300' <= actv <= '5750' then mixmail=1;
363      else if '1000' <= actv <= '5000' or actv in ('0350','0560')
then mixmail=0;
364      dollart=sum(of dollar1 - dollar4);
365      run;

```

NOTE: There were 1524 observations read from the data set WORK.OUT.

NOTE: The data set WORK.OUT has 1524 observations and 10 variables.

NOTE: DATA statement used (Total process time):

real time 0.00 seconds

cpu time 0.00 seconds

```

366
367      *-----;
368      * The following proc SUMMARY gets the preliminary results for ;
369      * Report 7 This corresponds to the old LIOCATT report ALA860P7;
370      *-----;
371      proc sort data=out out=out7;
372          by rgroup route actv;
373          where mixmail ne 1;
374      run;

```

NOTE: There were 1317 observations read from the data set WORK.OUT.

WHERE mixmail not = 1;

NOTE: The data set WORK.OUT7 has 1317 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time 0.00 seconds

cpu time 0.00 seconds

```

375
376      proc summary data=out7 noprint;
377          var dollar1 dollar2 dollar3 dollar4;
378          output out=rpt7 sum=;
379          by rgroup route actv;
380          /****uncomment print if report is needed***
381          proc print data=rpt7;
382              id rgroup route;
383              * sum dollar1 dollar2 dollar3 dollar4;
384              by rgroup route;
385              title2 'Report 7';
386              *****/
387      run;

```

NOTE: There were 1317 observations read from the data set WORK.OUT7.

NOTE: The data set WORK.RPT7 has 292 observations and 9 variables.

NOTE: PROCEDURE SUMMARY used (Total process time):

real time 0.00 seconds

cpu time 0.00 seconds

388

```

389 *****;
390 * Distribute mixed mail costs to direct mail activity codes. ;
391 *****;
392
393 *-----;
394 * Read the mixed-mail to direct mail code conversion file. ;
395 * This file tells which direct mail activity codes to associate;
396 * with each mixed-mail activity code. ;
397 *-----;
398 ;
399 Data mxmail;
400 set MxMailMap;
401 run;

```

NOTE: There were 311 observations read from the data set WORK.MXMAILMAP.

NOTE: The data set WORK.MXMAIL has 311 observations and 2 variables.

NOTE: DATA statement used (Total process time):

```

real time      0.01 seconds
cpu time       0.00 seconds

```

```

402
403 proc sort data=mxmail; by dircode;
404 run;

```

NOTE: There were 311 observations read from the data set WORK.MXMAIL.

NOTE: The data set WORK.MXMAIL has 311 observations and 2 variables.

NOTE: PROCEDURE SORT used (Total process time):

```

real time      0.00 seconds
cpu time       0.00 seconds

```

```

405
406 data dirmail (drop= i mixcode);
407     array dir(8) $mix1 -mix8;
408     do i=1 to 8;
409         set mxmail;
410         by dircode;
411         actv=dircode;
412         dir(i)=mixcode;
413         if last.dircode then i=8;
414     end;
415     output;
416 run;

```

NOTE: There were 311 observations read from the data set WORK.MXMAIL.

NOTE: The data set WORK.DIRMAIL has 107 observations and 10 variables.

NOTE: DATA statement used (Total process time):

```

real time      0.01 seconds
cpu time       0.00 seconds

```

```

417 *-----;
418 * Direct mail cost summary by cagfin, route and activity ;

```

```

419      *-----;
420  proc sort data=out out=dist2;
421      by cagfin route actv;
422      where mixmail=0; * Contains only direct mail;
423  run;

```

NOTE: There were 897 observations read from the data set WORK.OUT.

WHERE mixmail=0;

NOTE: The data set WORK.DIST2 has 897 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time 0.01 seconds

cpu time 0.00 seconds

```

424
425  proc summary data=dist2 noprint;
426      var dollar1 dollar2 dollar3 dollar4 dollart;
427      output out=dirdol
428          sum=dollar1d dollar2d dollar3d dollar4d dollartd;
429      by cagfin route actv;
430      title2 'dirdol';
431      * proc print data=dirdol;
432  run;

```

NOTE: There were 897 observations read from the data set WORK.DIST2.

NOTE: The data set WORK.DIRDOL has 897 observations and 10 variables.

NOTE: PROCEDURE SUMMARY used (Total process time):

real time 0.01 seconds

cpu time 0.00 seconds

```

433
434      *-----;
435      * Merge direct mail cost with the conversion table to associate;
436      * each direct mail activity code with mixed mail code that will;
437      * get distribued cost.
438      *-----;
439  proc sort data=dirdol;
440      by actv;
441  run;

```

NOTE: There were 897 observations read from the data set WORK.DIRDOL.

NOTE: The data set WORK.DIRDOL has 897 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time 0.00 seconds

cpu time 0.00 seconds

```

442  proc sort data=dirmail;
443      by actv;
444  run;

```

NOTE: There were 107 observations read from the data set WORK.DIRMAIL.

NOTE: The data set WORK.DIRMAIL has 107 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time	0.00 seconds
cpu time	0.00 seconds

```
445
446 data dircode1 (drop=mix1-mix8 i);
447     merge dirdol  (in=a)
448           dirmail (in=b keep=actv mix1-mix8);
449     by actv;
450     source=a+2*b;
451     if source=2 then delete; * delete activity codes that
contain
452                                     no cost;
453     array dir(8) mix1-mix8;
454     do i=1 to 8;
455         mixcode=dir(i);
456         if mixcode ne ' ' then output;
457         if mixcode= ' ' then i=8;
458     end;
459     *proc print n;
460 run;
```

NOTE: There were 897 observations read from the data set WORK.DIRDOL.

NOTE: There were 107 observations read from the data set WORK.DIRMAIL.

NOTE: The data set WORK.DIRCODE1 has 2837 observations and 12 variables.

NOTE: DATA statement used (Total process time):

real time	0.01 seconds
cpu time	0.00 seconds

```
461
462
463 *-----;
464 * Mixed mail cost summary by cagfin, route and actv      ;
465 *-----;
466
467 proc sort data=out out=dist1;
468     by cagfin route actv;
469     where mixmail=1; * Contains only mixed mails;
470 run;
```

NOTE: There were 207 observations read from the data set WORK.OUT.

WHERE mixmail=1;

NOTE: The data set WORK.DIST1 has 207 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time	0.01 seconds
cpu time	0.00 seconds

```
471
472 proc summary data=dist1 noprint;
473     var dollar1 dollar2 dollar3 dollar4 dollart;
474     output out=mxdol
```

```

475          sum=dollar1i dollar2i dollar3i dollar4i dollarti;
476          by cagfin route actv;
477          title2 'mxdol';
478          run;

```

NOTE: There were 207 observations read from the data set WORK.DIST1.

NOTE: The data set WORK.MXDOL has 207 observations and 10 variables.

NOTE: PROCEDURE SUMMARY used (Total process time):

```

      real time          0.00 seconds
      cpu time           0.00 seconds

```

```

479
480      *-----;
481      * Attach summary mixed mail cost to the direct mail activity  ;
482      * code.                                                         ;
483      *-----;
484
485      data mxdol;
486          set mxdol;
487          rename actv=mixcode;
488          *proc print data=mxdol;
489          run;

```

NOTE: There were 207 observations read from the data set WORK.MXDOL.

NOTE: The data set WORK.MXDOL has 207 observations and 10 variables.

NOTE: DATA statement used (Total process time):

```

      real time          0.01 seconds
      cpu time           0.00 seconds

```

```

490
491      proc sort data=dircode1;
492          by cagfin route mixcode;
493          run;

```

NOTE: There were 2837 observations read from the data set WORK.DIRCODE1.

NOTE: The data set WORK.DIRCODE1 has 2837 observations and 12 variables.

NOTE: PROCEDURE SORT used (Total process time):

```

      real time          0.00 seconds
      cpu time           0.00 seconds

```

```

494      proc sort data=mxdol;
495          by cagfin route mixcode;
496          run;

```

NOTE: There were 207 observations read from the data set WORK.MXDOL.

NOTE: The data set WORK.MXDOL has 207 observations and 10 variables.

NOTE: PROCEDURE SORT used (Total process time):

```

      real time          0.00 seconds
      cpu time           0.00 seconds

```

```

497
498 data final ;
499     merge dircode1 (in=a) mxdol (in=b);
500     by cagfin route mixcode;
501     source=a+2*b;
502
503     * proc print;
504     * title2 'final';
505     run;

```

NOTE: There were 2837 observations read from the data set WORK.DIRCODE1.

NOTE: There were 207 observations read from the data set WORK.MXDOL.

NOTE: The data set WORK.FINAL has 2837 observations and 17 variables.

NOTE: DATA statement used (Total process time):

real time 0.01 seconds

cpu time 0.00 seconds

```

506
507 *-----;
508 * Calculate the total direct mail cost associated with ;
509 * each mixed mail code. ;
510 *-----;
511
512 proc sort data=final;
513     by cagfin route mixcode;
514     run;

```

NOTE: There were 2837 observations read from the data set WORK.FINAL.

NOTE: The data set WORK.FINAL has 2837 observations and 17 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time 0.01 seconds

cpu time 0.00 seconds

```

515
516 proc summary data=final noprint;
517     var dollar1d dollar2d dollar3d dollar4d dollartd;
518     output out=total1
519         sum=total1d total2d total3d total4d totaltd;
520     by cagfin route mixcode;
521     run;

```

NOTE: There were 2837 observations read from the data set WORK.FINAL.

NOTE: The data set WORK.TOTAL1 has 413 observations and 10 variables.

NOTE: PROCEDURE SUMMARY used (Total process time):

real time 0.03 seconds

cpu time 0.00 seconds

```

522
523 *-----;
524 * Distribute the mixed mail cost to direct mail activity codes ;
525 *-----;

```

```

526         data distfile (drop=_type_ _freq_ source);
527         merge final (in=a) total1 (in=b);
528         by cagfin route mixcode;
529
530         if total1d=0 then ratio1=.;
531         else ratio1= dollar1d/total1d;
532         dist_1=dollar1i *ratio1;
533
534         if total2d=0 then ratio2=.;
535         else ratio2= dollar2d/total2d;
536         dist_2=dollar2i *ratio2;
537
538         if total3d=0 then ratio3=.;
539         else ratio3= dollar3d/total3d;
540         dist_3=dollar3i *ratio3;
541
542     *** The following statements add the undistributed dollar
543     *** for basic functions 1 - 3 to last basic function;
544
545         if (total1d =0 and dollar1i > 0)
546         then dollar4i=dollar4i +dollar1i;
547         if (total2d =0 and dollar2i > 0)
548         then dollar4i=dollar4i +dollar2i;
549         if (total3d =0 and dollar3i > 0)
550         then dollar4i=dollar4i +dollar3i;
551
552     *** Note that the last basic function distribution is based on
553     *** the total cost (sum of four basic functions cost);
554
555         if totaltd=0 then ratiot=.;
556         else ratiot= dollartd/totaltd;
557         dist_4=dollar4i *ratiot;
558
559     *** The following code assigns the undistributed mixed
560     *** dollars to the dist_0 cagegory;
561
562         if actv=' ' then
563         do;
564             actv=mixcode;
565             dist_0=dollarti;
566         end;
567
568     *Use macro to assign route group;
569     %assignRouteGroup;
MPRINT(ASSIGNROUTEGROUP):    *** The following statement assigns route
code groups *** so that
reports can be created separately for *** these groups.;
MPRINT(ASSIGNROUTEGROUP):    *set rgroup for special runs, for delivery
cost model (3 values);
MPRINT(ASSIGNROUTEGROUP):    *Put Letter Route types in RGroup 1, SPR in
RGroup 2, 99 in RGroup 3;
MPRINT(ASSIGNROUTEGROUP):    if '71' <= route <= '83' then RGroup=1;
MPRINT(ASSIGNROUTEGROUP):    else if '84' <= route <= '98' then RGroup=2;
MPRINT(ASSIGNROUTEGROUP):    else RGroup=3;

```

```

570
571
572      * proc print data=distfile n;
573 run;

```

NOTE: Missing values were generated as a result of performing an operation on missing values.

Each place is given by: (Number of times) at (Line):(Column).

2748 at 532:27 659 at 536:27 2837 at 540:27 659 at 557:27

NOTE: There were 2837 observations read from the data set WORK.FINAL.

NOTE: There were 413 observations read from the data set WORK.TOTAL1.

NOTE: The data set WORK.DISTFILE has 2837 observations and 29 variables.

NOTE: DATA statement used (Total process time):

```

real time          0.04 seconds
cpu time           0.00 seconds

```

```

574
575      *-----;
576      * Combine schedule K and L to produce K and L report      ;
577      * (report 13). Schedule K is report 7 and schedule L is    ;
578      * summary of the mixed mail distribution result (finout).  ;
579      *-----;
580
581      proc sort data=distfile;
582          by   rgroup route actv;
583      run;

```

NOTE: There were 2837 observations read from the data set WORK.DISTFILE.

NOTE: The data set WORK.DISTFILE has 2837 observations and 29 variables.

NOTE: PROCEDURE SORT used (Total process time):

```

real time          0.00 seconds
cpu time           0.00 seconds

```

```

584
585      proc summary data=distfile noprint;
586          var   dist_0 dist_1 dist_2 dist_3 dist_4;
587          output out=finoutl sum=dollar0 dollar1 dollar2 dollar3
dollar4;
588          by   rgroup route actv;
589      run;

```

NOTE: There were 2837 observations read from the data set WORK.DISTFILE.

NOTE: The data set WORK.FINOUTL has 202 observations and 10 variables.

NOTE: PROCEDURE SUMMARY used (Total process time):

```

real time          0.02 seconds
cpu time           0.00 seconds

```

```

590
591      *-----;
592      * The following proc format classifies activity codes into ;
593      * categories for subsequent reports.                        ;

```

```

594  *-----;
595  proc format ;
596  value $actv
597  /*SPECIAL SERVICES */
598      '0060'          ='055 Registered  '
599      '0050'          ='051 Certified   '
600      '0080'          ='054 Insured    '
601      '0030'          ='052 COD        '
602
603  /* OTHSERV INCL. BUSRPLY, RETRCPT, ADDRRCORR, FORMS3547&3579 */
604      '6020','6030'='074 PO Box      '
605      '0090','0190',
606      '0300','0100',
607      '0110','0140'          ='058 Other Services'
608
609  /* FIRST CLASS */
610      '1060'          ='003 FC SP Letters  '
611      '1020'          ='004 FC SP Cards    '
612      '1080','1081','1085','1086'='008 FC Prst Letters'
613      '1022','1035','1040','1045'='009 FC Prst Cards  '
614      '2060'          ='014 FC SP Flats    '
615      '2080','2081','2086'      ='017 FC Prst Flats  '
616      '3060','4060'          ='143 FC Package Svc '
617      /*'3080','4080'          ='020 FC Prst Parcels'*/
618      '3080','4080'          ='143 FC Package Svc '
619
620
621  /*PRIORITY MAIL */
622      '1160','2160','3160','4160'='148 Priority      '
623
624  /* EXPRESS MAIL */
625      '1110','2110','3110','4110'='140 Express      '
626
627  /* PERIODICALS */
628      '1211','2211','3211','4211'='031 PER In County  '
629      '1212','2212','3212','4212'='032 PER Outside County'
630
631  /* STANDARD */
632      '1317'          ='021 MM ECR HD Letters
633      '2317','3317','4317'      ='022 MM ECR HD Flats/Parcels
634      '1318'          ='021 MM ECR SAT Letters
635      '2318','3318','4318'      ='022 MM ECR SAT
636  Flats/Parcels  '
637      '1312','2312','3312','4312'='023 MM ECR Carrier Route
638      '1320','2320','3320','4320'='024 MM EDDM - Retail
639      '1340','1341','1345'      ='025 MM REG Letters
640      '2340','2341','2345'      ='026 MM REG Flats

```

```

640          '3340','4340'          ='027 MM REG Parcel
641          '3341','4341'          ='027 MM REG Parcel
642          '1315'                  ='021 DAL Ltr WSH
643          '2315','3315','4315'    ='021 DAL Flt/IPP/PAR WSH
644          '1316'                  ='022 DAL Ltr WSS
645          '2316','3316','4316'    ='022 DAL Flt/IPP/PAR WSS
646          '1319','2319','3319','4319'='021 Parent No Dal HD/SAT
647
648  /* PACKAGE          */
649          '2402','3402','4402'      ='157 Ground Advantage HW 1-
31b '
650          '2403','3403','4403'      ='158 Ground Advantage HW
over31b '
651          '2405','3405','4405'      ='156 Ground Advantage LW
652          '2410','3410','4410'      ='145 PKG Retail Ground
653          '2460','2465','2480','2495' ='042 PKG BPM Flats
654          '3460','3465','3480','3495',
655          '4460','4465','4480','4495' ='043 PKG BPM Parcels
656          '1420','1425','1430',
657          '2420','2425','2430',
658          '3420','3425','3430',
659          '4420','4425','4430'      ='044 PKG Media
660
661  /* PARCEL SELECT and PS Lightweight          */
662          '2360','3360','4360',
663          '2493','3493','4493'      ='151 PKG Parcel Select
664
665  /* PARCEL RETURN SERVICE (PRS)          */
666          '2475','3475','4475'      ='154 PKG Parcel Return (PRS)
667
668  /* PREMIUM FORWARDING SERVICE (PFS)          */
669          '0350','1170','2170','3170','4170'='170 Prem. Forw. Svc'
670
671
672  /* GOVERNMENT MAIL          */
673          '0560',
674          '1510','2510','3510','4510'='085 USPS
675
676  /* FREE          */
677          '1910','2910','3910','4910'='086 Free

```

```

678
679  /* INTERNATIONAL OUTBOUND AND INBOUND */
680      '1780','2780','3780','4780',
681      '5081','6081','5082','6082',
682      '0752','0757','0762','0767',
683      '0770','0860','0862','0867'           ='185
International'
684
685  /* OTHER CODES */
686      OTHER = '999 OTHER';
NOTE: Format $ACTV is already on the library WORK.FORMATS.
NOTE: Format $ACTV has been output.
687      run;

```

```

NOTE: PROCEDURE FORMAT used (Total process time):
      real time           0.04 seconds
      cpu time            0.00 seconds

```

```

688
689  data KL (drop=_type_ _freq_);
690      set rpt7 (in=a) finout1 (in=b);
691      if a then source='K';
692      if b then source='L';
693      if dollar0 > 0 then
694          do; source='U';
695              dollar4=dollar0; *move the undistributed mixed
mail
696                          cost to basic function OTHER;
697          end;
698
699      if dollar0=. and dollar1=. and dollar2=. and dollar3=. and
700      dollar4=. then delete;
701
702      total= sum (of dollar1 - dollar4);
703      if actv <= '0900' and actv not in ('0350','0560')
then actgrp=1;
704      else if '1020' <= actv <= '4910' or actv in
('0350','0560') then actgrp=2;
705      else if '5300' <= actv <= '5750' then actgrp=3;
706      else if '5020' <= actv <= '6720' then actgrp=4;**bls;
707      **use format to classify activity to classes**;
```

708	class=put(actv,\$actv.);
709	
710	**assign labels for use in reports**;
711	label
712	dollar1 = 'OUTGOING'
713	dollar2 = 'INCOMING'
714	dollar3 = 'TRANSIT '
715	dollar4 = 'OTHER ';
716	
717	**restrict to letter routes only**;
718	*if rgroup=1;
719	run;

NOTE: There were 292 observations read from the data set WORK.RPT7.
NOTE: There were 202 observations read from the data set WORK.FINOUTL.
NOTE: The data set WORK.KL has 493 observations and 12 variables.
NOTE: DATA statement used (Total process time):
 real time 0.03 seconds
 cpu time 0.00 seconds

```
720
721
722 *Generate class variable from lookup table;
723 *import craft,activity code combinations used in delivery cost
model;
724 PROC IMPORT OUT= WORK.ActivityCodesTable
725             DATAFILE=
"&pathprog\ActivityCodesMaster_ForIndicia.xlsx"
SYMBOLGEN: Macro variable PATHPROG resolves to D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles
726             DBMS=xlsx REPLACE;
727 RUN;
```

NOTE: One or more variables were converted because the data type is not supported by the V9

 engine. For more details, run with options MSGLEVEL=I.
NOTE: The import data set has 157 observations and 3 variables.
NOTE: WORK.ACTIVITYCODESTABLE data set was successfully created.
NOTE: PROCEDURE IMPORT used (Total process time):
 real time 0.03 seconds
 cpu time 0.00 seconds

```
728 proc sql;
729 create table KL2 as
730     select KL.*, ac.NewCRAProd as class2
731     from KL left join ActivityCodesTable as ac
732     on KL.actv = ac.ActivityCode;
NOTE: Table WORK.KL2 created, with 493 rows and 13 columns.
```

```
733 quit;
NOTE: PROCEDURE SQL used (Total process time):
    real time                0.00 seconds
    cpu time                0.00 seconds
```

```
734
735 data checkClass; set KL2;
736 where class ne class2;
737 run;
```

NOTE: There were 182 observations read from the data set WORK.KL2.
 WHERE class not = class2;
NOTE: The data set WORK.CHECKCLASS has 182 observations and 13 variables.
NOTE: DATA statement used (Total process time):

```
real time          0.00 seconds
cpu time           0.00 seconds
```

```
738
739 ***zzz;
740
741 data KL; set KL2;
742 if class2 = "" then class2 = "999 OTHER";
743 if substr(class2,1,3) = "185" then class = "185 International";
744     else class = class2;
745 drop class2;
746 run;
```

NOTE: There were 493 observations read from the data set WORK.KL2.

NOTE: The data set WORK.KL has 493 observations and 12 variables.

NOTE: DATA statement used (Total process time):

```
real time          0.00 seconds
cpu time           0.00 seconds
```

```
747 Data kl; set kl;
748 If actv = '1060' or actv = '1061' or actv = '1062' or actv = '1063'
or actv = '1064' or actv
748! = '1065' or actv = '1066';
749
750 *Export KL table by route, for delivery cost model;
```

NOTE: There were 493 observations read from the data set WORK.KL.

NOTE: The data set WORK.KL has 18 observations and 12 variables.

NOTE: DATA statement used (Total process time):

```
real time          0.01 seconds
cpu time           0.00 seconds
```

```
751 proc export data=kl
752     outfile = "&pathOut\KLTable_FCSP_Letter_Collection&FYData..xlsx"
SYMBOLGEN:  Macro variable PATHOUT resolves to D:\ExternalSSD\Folder
19\FY25\SAS\KLTables
SYMBOLGEN:  Macro variable FYDATA resolves to 25
753     dbms = xlsx replace;
754 run;
```

NOTE: The export data set has 18 observations and 12 variables.

NOTE: "D:\ExternalSSD\Folder

19\FY25\SAS\KLTables\KLTable_FCSP_Letter_Collection25.xlsx" file was
successfully created.

NOTE: PROCEDURE EXPORT used (Total process time):

```
real time          0.04 seconds
cpu time           0.00 seconds
```